Two-Period Macroeconomic Model

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Introduction

- Beyond a “hand-to-mouth” economy where everything produced today must be consumed today.
- The consumption-savings choice revisited.
- The real interest rate as the relative price of goods today v. goods tomorrow.
- Macro implications: Understanding how expectations of the future (taxes, income, productivity) affect current economic activity.
- Today: Perfect foresight.

Summary

Consumers

Firms

Revisiting the Market of Loanable Funds

Government

Equilibrium

Consumers

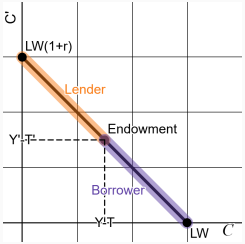
Per-Period Budget Constraints

$$\begin{cases} C + S = Y - T \\ C' = Y' - T' + (1 + r)S \end{cases}$$

- In the initial period the consumer splits her disposable income into consumption and savings.
- In the final period the consumer uses all her disposable income plus savings from the initial period.

Intertemporal Budget Constraint

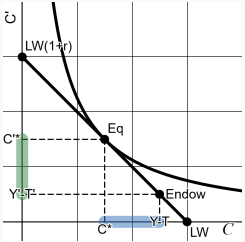
$$C + \frac{C'}{1+r} = Y - T + \frac{Y' - T'}{1+r}$$



- Lifetime budget constraint.
- Present value.
- Relative price.
- Lifetime wealth.
- Compare with $p_1c_1 + p_2c_2 = m$.

Optimal Choice

$$\text{MRS}_{C,C'} = 1 + r$$



- Tangent indifference curve.
- Lender case in the picture.
- Borrower?
- Savings and return to savings in the graph.
- Consumption smoothing.

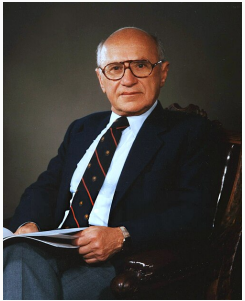
Increase in Current Income

- The lifetime budget constraint shifts out, with the slope unchanged.
- Current and future consumption both increase (they are both normal goods).
- Current consumption increases by less than the increase in current income.
- The consumer saves/borrows part of the increase in current income.

Increase in Future Income

- The lifetime budget constraint shifts out, with the slope unchanged.
- Current and future consumption both increase (they are both normal goods).
- Future consumption increases by less than the increase in future income.
- The consumer saves less.

The Permanent Income Hypothesis



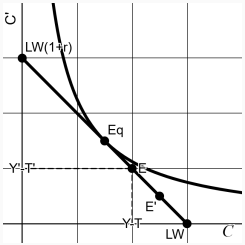
Milton Friedman

- When a consumer receives a change in her current income, it matters for her choice whether this change in income is temporary or permanent.
  - Temporary increases in income increase current consumption by a small amount.
  - Permanent increases in income increase consumption by a large amount.
- The primary determinant of a consumer's current consumption is her permanent income.

Increase in the Interest Rate

- Budget constraint pivots.
- Two effects:
  1. Substitution effect: savings have a higher return. Then the agent consume less today and saves more.
  2. Income effect: higher income in the future. Then the agent wants to consume more (today and in the future).
- Substitution effects pushes towards more savings, but income effects towards less savings.
- Overall effect is undetermined.

Ricardian Equivalence



- Assume the government cut taxes today, but increase them in the future. Equivalent amount.
- Intertemporal budget constraint stays the same. Optimal choice doesn't change.
- Consumer saves up for future taxes.
- Assumption: can perfectly save/borrow.

The Market for Loanable Funds

- Consumers with different endowments are enough to understand how this market works.
- Lenders have extra income today and are willing to lend it for a real reward in the future.
- Borrowers expect to have a good income in the future and are willing to borrow today and pay an extra in the future.
- Example: students and adults.
- More agents: firms and government.

Firms

Current & Future Production

$$Y = A \cdot F(K, L)$$
$$Y' = A' \cdot F(K', L')$$

- Assume K is given. So the firm chooses:
  - Labor input today L.
  - Labor input tomorrow L'.
  - Investment in new capital.
- Goal maximize the (discounted) sum of profits.

Capital Accumulation Formula

$$K' = (1 - \delta)K + I$$

- The future capital input is the sum of the current capital input (net of depreciation) and gross investment I.
- Simple model of depreciation. Each period a fraction δ of current capital goes to waste.

Timing

- In the current period K is given.
- Choose L for current production and devote I to increase the capital stock next time.
- In the next period K' is determined by previous investment.
- Choose L' for future production.
- No need to devote any resource to new investment.

Marginal Cost of Investment

$$\mathrm{MC}(I) = 1$$

- What the firm gives up, in terms of the present value of profits, by investing one unit of capital in the current period.
- An additional unit of investment I reduces profits by 1.

Marginal Benefit of Investment

$$\mathrm{MB}(I) = \frac{\mathrm{MP}'_K + 1 - \delta}{1 + r}$$

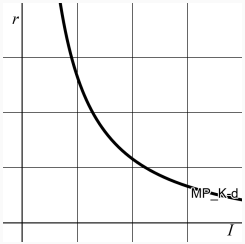
- What one extra unit of investment adds to the present value of profits.
- Future marginal product.
- Future stock of capital at the end of tomorrow.

Optimal Investment Decision

$$\mathrm{MP}'_K - \delta = r$$

- Firm invests until the net marginal product of capital is equal to the interest rate.
- The opp. cost of investing in more capital is the real interest rate.
- Defines the investment demand curve.

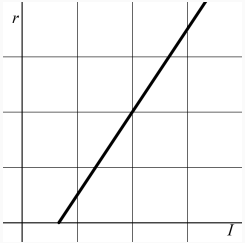
Investment Demand



- Negative slope due to the law of diminishing returns.
- It describes the optimal investment choice of the firm.
- Shifts to the right if:
  - Current capital decreases.
  - Future TFP increases.
  - Depreciation increases.

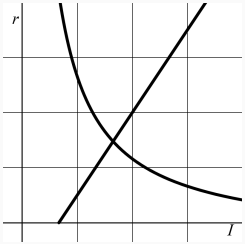
Revisiting the Market of Loanable Funds

Savings Supply



- Assume there's no government. So no taxes to consumers.
- Assume overall consumers are saving.
- Assume the substitution effect dominates.
- National savings are private savings.

Equilibrium



- Thrift and productivity determine the real interest rate.
- Changes in preferences or present/future income shift savings supply.
- Changes in productivity or present capital (or its depreciation) shift the investment demand.

Government

Per Period Budget Constraints

$$\begin{cases} G = T + B \\ G' + (1+r)B = T' \end{cases}$$

- Govn't finances current expenditures with taxes or bonds.
- In the future period uses future taxes to repay bonds and future expenditures.

Intertemporal Budget Constraint

$$G + \frac{G'}{1+r} = T + \frac{T'}{1+r}$$

- The present value of government purchases must equal the present value of taxes.
- The government must eventually pay off all of its debt by taxing its citizens.

Equilibrium

One-Period Model: Markets and Prices

1. Goods market.
   - Price is normalized $p = 1$.
   - Firms supply output Y, consumers and government demand C and G.
2. Labor market.
   - Price is the real wage w.
   - Firms demand labor and consumer supply it.

Two-Period Model: Markets and Prices

1. Goods market. Current and future.
   - Prices are normalized.
   - Firms supply Y (and Y'), consumers and government demand C and G (and C' and G').
2. Labor market. Current and future.
   - Prices are real wages: w and w'.
   - Firms demand labor and consumers supply labor.
3. Market of Loanable Funds.
   - Price is the real interest rate r.
   - Consumers (and government) save/borrow and firms invest.

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